

QUESTIONS TO REVIEW SINGLE-VARIABLE CALCULUS

1. Consider the piecewise function

$$f(x) = \begin{cases} \frac{1}{x} & x < 1 \\ x - 1 & x \geq 1 \end{cases}. \quad (1)$$

At what points do discontinuities occur, and what type of discontinuities occur at those points?

2. Suppose $f(x) = 1/x$. Over the interval $[a, b] = [-1, 1]$,

- (a) What is $[f(a), f(b)]$?
- (b) Find a value within $[f(a), f(b)]$ which $f(x)$ does not obtain over $[-1, 1]$.
- (c) Why does (b) not contradict the Intermediate Value Theorem?

3. Suppose the power output of a cyclist is $\mathcal{W}_0$ watts and is governed by the equation

$$\mathcal{W} = C_1 v^2, \quad \text{where } C_1 \text{ is a positive constant.} \quad (2)$$

If we wish to keep the cyclist's power output to within ε watts of $\mathcal{W}_0$, by how much can the velocity vary?

4. Here, we will show that $\frac{d}{dx}x^n = nx^{n-1}$. We start with the limit definition:

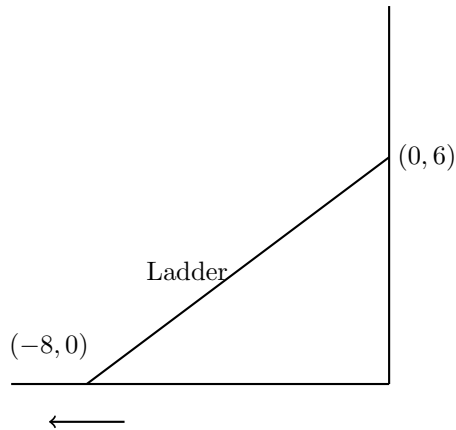
$$\frac{d}{dx}x^n = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}. \quad (3)$$

- (a) Use the binomial theorem on $(x+h)^n$ and expand it as a sum of terms.
- (b) Cancel out the x^n terms in the numerator and simplify the expression.
- (c) Factor out an h from the numerator and cancel it with the denominator.
- (d) Take the limit. What do you get? Verify that it is nx^{n-1} .

5. Using the limit definition of the derivative, derive the following derivatives:

- (a) $d \ln(x)/dx = 1/x$. *Hint: denote the limit as L and try manipulating the equation.*
- (b) $d \sin(x)/dx = \cos(x)$.
- (c) $dC^x/dx = C^x \ln C$. *Hint: you may want to expand something as its Taylor Series at some point, but this might also not be strictly necessary. That's just how I did it.*

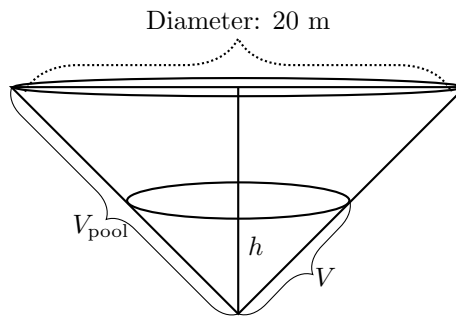
6. Suppose a 10 meter ladder slides down a wall.



Our ladder at time t .

Its horizontal velocity is 2 meters per second to the left at some time t . What is its vertical velocity at that time?

7. A conical swimming pool is being filled up with water. The pool's depth is 10 meters and the diameter at surface level is 20 meters. Water is being pumped into the pool at a rate of 20 liters per minute. When the pool is half filled *by volume*, how fast is the water level rising?



- Express the volume of the pool V in terms of the height and radius. Then find a relation such that V is a function of h **only** (ie. $V(r, h) \rightarrow V(h)$).
 - Perform implicit differentiation so you get dV/dt in terms of dh/dt .
 - Find the height, h , of the pool when it is half filled by volume. Then substitute all the values you have to find dh/dt . Be careful of your units!
8. The sigmoid function, used widely as an activation function in artificial neural networks, has the form

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \quad (4)$$

- Without evaluating $\sigma'(x)$ (i.e., taking the derivative of σ), state a value that $\sigma'(x)$ must take on within the interval $x \in [-\ln 2, \ln 2]$. Let this value be α . *Hint: Mean Value Theorem.*
- Evaluate $\sigma'(x)$, then find a c such that $\sigma'(c) = \alpha$. If there is more than one solution, state only one of them. Moreover, what is the maximum value that $\sigma'(x)$ obtains over all real numbers, and at what value of x is it obtained? *Hint: For the first part, you may find it helpful to write $\sigma'(x)$ such that it involves the hyperbolic cosine, $\cosh(x) = \frac{1}{2}(e^x + e^{-x})$.*

- (c) Using l'Hopital's rule, evaluate $\lim_{x \rightarrow -\infty} \sigma'(x)$. Why are we permitted to use l'Hopital's rule in this case?
- (d) Show that $\sigma(x)$ only has one inflection point, which occurs at $x = 0$.
- (e) Over the interval $(0, \infty)$, show that $\sigma(x)$ is concave down, meaning tangent lines drawn to the function consistently overestimate neighboring values of the function.
- (f) Derive the differential equation for which $\sigma(x)$ is a solution in the form $\sigma' = f(\sigma)$.
- (g) Sigmoid curves can also model population growth over time, if we make a change of variables $x \mapsto t$. With your differential equation from the previous question, interpret and explain the value of σ' in the following regimes, in the context of using σ to model a population:
- When σ is very small, $\sigma \ll 1$.
 - When $\sigma \sim 1/2$.
 - When σ is less than but close to 1.
 - When $\sigma > 1$.
9. Find the arc length of the function $\sin(x)$ over $x \in [0, 2\pi]$ (you may use computational software to evaluate the final integral).
10. Evaluate the following integrals.
- (a)
- $$I_1 = \int_0^\infty r e^{-r^2} dr. \quad (5)$$
- (b)
- $$I_2 = \int_{-\pi}^\pi x \sin(x) dx. \quad (6)$$
- (c)
- $$I_3 = \int_{-a}^a \frac{1}{2} (f(t) - f(-t)) dt. \quad (7)$$
- Note: the function $\frac{1}{2}(f(t) - f(-t))$ is sometimes called the antisymmetrization of f . See if you can deduce why.*
- (d)
- $$I_4 = \int_0^\pi \sin^2(x) dx. \quad (8)$$
- (e) The following integral, I_5 , is usually first encountered when finding the electric field some fixed distance R away from a charged, one-dimensional object, like a rod or ring:
- $$I_5 = \int_{-a}^a \frac{1}{(R^2 + x^2)^{3/2}} dx. \quad (9)$$
11. Show that the volume of a cone is $V = \frac{1}{3}\pi r^2 h$ via integration methods. *Hint: Disk method.*
12. Consider the polar function $r = \cos 2\theta$, which over $[-\pi, \pi]$, forms a structure resembling a four-leaf clover. Compute the area of one of these “petals” (you may graph in Desmos to visualize).
- Questions 13-15 deal with series.

13. Show that the series

$$S = \sum_{n=1}^{\infty} \frac{1}{n^p} \quad (10)$$

converges for $p \geq 2$. What is the limit of S as $p \rightarrow \infty$?

14. Find the first three nonzero terms of the Maclaurin Series for the following functions, and find closed form infinite series solutions for (b) and (d):

(a)

$$f_1(x) = \frac{1}{1-x^2}. \quad (11)$$

For this function, also determine its radius of convergence.

(b)

$$f_2(x) = \sin(x). \quad (12)$$

(c)

$$f_3(x) = \tan(x). \quad (13)$$

(d)

$$f_4(x) = \ln(1+x). \quad (14)$$

15. Derive the following identity:

$$e = \sum_{n=1}^{\infty} \frac{1}{n!}. \quad (15)$$

16. You may have encountered the linearized pendulum equation, which is a valid approximation for the full pendulum equation when the pendulum has small oscillations:

$$\ddot{\theta} + \frac{g}{\ell} \sin \theta = 0, \quad (\text{general})$$

$$\ddot{\theta} + \frac{g}{\ell} \theta = 0. \quad (\text{linearized})$$

As a reminder, n dots over a variable indicate its n th time derivative. For this question in particular, g is the gravitational acceleration and ℓ is the length of the string on which the mass is attached.

- (a) Over what regime is this “small angle approximation” a good approximation? In other words, at about what θ do higher order terms in the Taylor series of $\sin(\theta)$ start contributing non-negligibly? Assume the threshold for “non-negligible contributions” start when higher order terms correct the linearized result by more than 10%.
- (b) Solving the general pendulum equation gives us a robust model for oscillations of all amplitudes. We’ll start this process by using something called an *integration factor*, which in this case amounts to multiplying both sides of the general equation by $\dot{\theta}$:

$$\dot{\theta} \ddot{\theta} + \frac{g}{\ell} \dot{\theta} \sin \theta = 0. \quad (16)$$

Rewrite equation 16 such that the left side is the time derivative of some function of θ and $\dot{\theta}$; that is, rewrite the equation into the form $\frac{d}{dt} f(\theta, \dot{\theta}) = 0$, which allows us to deduce that $f(\theta, \dot{\theta})$ must be a constant. We will call this constant α for now.

- (c) Let's think about the system physically. We assume that when $t = 0$, the pendulum is at its maximum angular displacement and at zero angular velocity. If we denote the initial/maximum angular displacement as θ_0 , given the above, deduce that $\alpha = -\frac{g}{\ell} \cos \theta_0$.
- (d) Now that we have deduced α , show that, once simplified, we obtain the following first-order separable differential equation:

$$\dot{\theta} = \sqrt{\frac{2g}{\ell} (\cos \theta - \cos \theta_0)}. \quad (17)$$

- (e) If we invoke the small angle approximation here, show that we retrieve the simple harmonic oscillator solution, as expected. In particular, show that

$$\theta(t) = A \sin(\omega t + \delta), \quad \omega = \sqrt{\frac{g}{\ell}}, \quad \delta = \sin^{-1} \left(\frac{\theta_0}{A} \right), \quad A = \sqrt{2}(1 - \cos \theta_0). \quad (18)$$

- (f) From our result in (d), one can, in principle, just separate and integrate to find the full $\theta(t)$, although the integral itself cannot be evaluated with elementary functions. However, we can show that the period of a pendulum is not simply $T = 2\pi\sqrt{\ell/g}$, as is seen in introductory physics textbooks, but has nuance simply neglected in the small angle approximation. Let's illustrate how this is.

First, we recognize that the pendulum oscillates symmetrically, meaning that the time it takes for the pendulum to travel from θ_0 to 0 is the same as from 0 to θ_0 . This rather simple observation lets us conclude that the total period of our pendulum, T , is four times as long as the time between the pendulum being at its maximal angular displacement θ_0 and at zero angular displacement. This observation will help us determine bounds of integration.

Next, as you did in (e), separate the differential equation but do not apply the small angle approximation. Set up the integrals and show that

$$\int_0^{\theta_0} \left(\frac{2g}{\ell} (\cos \theta - \cos \theta_0) \right)^{-1/2} d\theta = \int_0^{T/4} dt. \quad (19)$$

- (g) Taking out the constant factors on the left hand side and integrating the right hand side, we have

$$T = \frac{2\sqrt{2}}{\omega} \int_0^{\theta_0} \frac{d\theta}{\sqrt{\cos \theta - \cos \theta_0}}, \quad \omega = \sqrt{\frac{g}{\ell}}. \quad (20)$$

Employ half angle identities and first show that

$$T = \frac{2}{\omega} \int_0^{\theta_0} \frac{d\theta}{\sqrt{\sin^2(\theta_0/2) - \sin^2(\theta/2)}}. \quad (21)$$

Define $k = \sin(\theta_0/2)$ and make an appropriate u -substitution from θ to ϕ to derive

$$T = \frac{4}{\omega} K(k), \quad K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}. \quad (22)$$

The function $K(k)$ is known as the *complete elliptic integral of the first kind*.

- (h) Taylor expand the integrand in $K(k)$ to three terms, truncating higher order terms, and integrate. Use computational software to evaluate the integrals in ϕ (unless you want to do them by hand!). Once you've done this, Taylor expand k^2 with respect to θ_0 and keep only the first two nonzero terms, then expand and group together terms sharing the same degree in θ_0 , truncating the series after the third nonzero term. Do you see the familiar $T = 2\pi\sqrt{\ell/g}$? Plot the deviation due to the two higher order terms you kept. How large is the fractional deviation at $\theta_0 = 1^\circ, 30^\circ, 90^\circ$?